MAL3040 — Statistics
Assignment 2: Sampling Distributions

Department of Mathematics, Indian Institute of Technology Jodhpur

B.Sc.–B.Ed. III year • Issued: 5 September 2026 • Due: 7 September 2026

About this sheet. There are eleven questions. Part A asks you to prove two facts that the tables do not print. Part B is about reading the tables. Part C is about the distributions themselves. You will need Tables A1 to A4 from the appendix of Ross, and a calculator.

Please note. The observations in Question 8 were made up for practice. They are not real measurements.

What you will need. You have a random sample $X_1, X_2, \ldots, X_n$ from a population with mean μ and variance σ^2. This means that the X_i are independent, and that all of them have the same distribution.

- Sample mean: $\overline{X} = \dfrac{1}{n}\sum_{i=1}^{n} X_i$.
- Sample variance: $S^2 = \dfrac{1}{n-1}\sum_{i=1}^{n}\big(X_i - \overline{X}\big)^2$. Always divide by $n-1$, never by n.
- Z is always a standard normal random variable, and Φ is its distribution function.
- *Upper tail points.* Let T have the t distribution with k degrees of freedom. Then $t_{\alpha,k}$ is the number for which $P(T > t_{\alpha,k}) = \alpha$. The symbols $\chi^2_{\alpha,k}$ and $F_{\alpha,m,k}$ have the same meaning. In $F_{\alpha,m,k}$, the first number m is the degrees of freedom of the numerator.
- *The χ^2 distribution.* χ^2_k is the distribution of $Z_1^2 + \cdots + Z_k^2$, where $Z_1, \ldots, Z_k$ are independent standard normal random variables.
- *The t distribution.* Let $Z \sim N(0,1)$ and $V \sim \chi^2_k$ be independent. The t distribution with k degrees of freedom is the distribution of $Z/\sqrt{V/k}$.
- *The F distribution.* Let $V_1 \sim \chi^2_m$ and $V_2 \sim \chi^2_k$ be independent. The F distribution with m and k degrees of freedom is the distribution of $(V_1/m)\big/(V_2/k)$.

You will not need the density of any of these distributions on this sheet.

Part A. Two identities that are not printed in the tables

1. Let $Z \sim N(0,1)$ and $V \sim \chi^2_k$ be independent. Put $T = Z/\sqrt{V/k}$, so that T has the t distribution with k degrees of freedom.

(a) Show that $-T$ has the same distribution as T. (Hint: What is the distribution of $-Z$? Is $-Z$ still independent of V?)

(b) Use part (a) to show that $P(T \leq -t) = P(T \geq t)$ for every real number t.

(c) Now show that $t_{1-\alpha,\,k} = -\,t_{\alpha,\,k}$ for every $0 < \alpha < 1$. Why does the table then need to give only the values of α that are less than $1/2$?

2. Let $V_1 \sim \chi^2_m$ and $V_2 \sim \chi^2_k$ be independent. Put $W = \dfrac{V_1/m}{V_2/k}$, so that W has the F distribution with m and k degrees of freedom.

(a) Show that $1/W$ has the F distribution with k and m degrees of freedom. Note that the two numbers change places.

(b) Let $c = F_{1-\alpha, m,k}$. Then $P(W > c) = 1 - \alpha$, and so $P(W \leq c) = \alpha$. Write the event $\{W \leq c\}$ as an event about $1/W$, and show that

$$F_{1-\alpha, m,k} = \frac{1}{F_{\alpha, k,m}}.$$

(c) The F table gives only small values of α. Explain in one or two sentences why part (b) is needed to answer a question about the *lower* tail of an F distribution. Why is there no such identity for the χ^2 distribution?

Part B. Using the tables

3. Use the table of Φ to find the following. Whenever the number is negative, say which property of the normal distribution you have used.

(a) $P(Z \leq 1.27)$

(b) $P(Z > 0.58)$

(c) $P(-1.44 \leq Z \leq 2.05)$

(d) $P(Z \leq -2.33)$

(e) $P(|Z| \leq 1.96)$

(f) Among the entries of the table, find the value of z for which $P(Z > z)$ is closest to 0.10.

4. Let T have the t distribution with 15 degrees of freedom.

(a) Write down $t_{0.05,\,15}$ and $t_{0.025,\,15}$.

(b) Find $P(T > 2.131)$ and $P(T < 2.602)$.

(c) Use Question 1(c) to find the number a for which $P(T < a) = 0.05$. This number is negative, so you cannot read it from the table directly.

(d) Find the number b for which $P(|T| > b) = 0.05$.

(e) The table gives the degrees of freedom up to 29, and then a last row marked ∞. Write down $t_{0.10,\,29}$, and also the entry in the same column of the row marked ∞. Compare both of these with the value of z that you found in Question 3(f). What do you see? What does this tell you about the t distribution when the number of degrees of freedom is large?

5. Let V have the χ^2 distribution with 10 degrees of freedom.

(a) Write down $\chi^2_{0.05,\,10}$ and $\chi^2_{0.01,\,10}$.

(b) Find $P(V > 20.483)$.

(c) Find two numbers a and b for which $P(V < a) = 0.025$ and $P(V > b) = 0.025$. Then $P(a \leq V \leq b) = 0.95$.

(d) In part (c) you had to look up two entries of the table. Explain why you cannot get b from a, as you could for the t distribution in Question 4(c).

6. Let W have the F distribution with 4 degrees of freedom for the numerator and 5 for the denominator. In the table, the degrees of freedom of the numerator are along the top, and the degrees of freedom of the denominator are down the side.

(a) Write down $F_{0.05,\,4,5}$, and find $P(W > 5.19)$.

(b) Write down $F_{0.05,\,5,4}$. Why is it different from your answer in part (a)?

(c) Use Question 2(b) to find the number c for which $P(W < c) = 0.05$. Say clearly which entry of the table you have used, and which degrees of freedom you took in each position.

(d) The F table gives only one value of α, namely $\alpha = 0.05$. Which parts of this question could you not have answered without the identity of Question 2?

Part C. Working with the distributions

7. A random variable X has the *Bernoulli distribution with parameter* p, where $0 \le p \le 1$, if $P(X = 1) = p$ and $P(X = 0) = 1 - p$.

(a) Find $E[X]$.

(b) Note that $X^2 = X$. Use this to find $E[X^2]$, and then show that $\mathrm{Var}(X) = p(1-p)$.

(c) For which value of p is the variance largest? Complete the square to give a reason.

(d) Let $X_1, \dots, X_n$ be a random sample from this distribution, and let $Y = X_1 + \cdots + X_n$. Then Y has the *binomial distribution* with parameters n and p. Use the results proved in class for $\overline{X}$ to show that $E[Y] = np$ and $\mathrm{Var}(Y) = np(1-p)$. Do not sum any binomial series.

(e) Explain why $\overline{X} = Y/n$ is the proportion of ones in the sample. Write down $E[\overline{X}]$ and $\mathrm{Var}(\overline{X})$.

(f) In part (d), at which step did you use the independence of the X_i? At which step did you use only the fact that they have the same distribution?

8. Recall from class that

$$\sum_{i=1}^{n} (X_i - \overline{X})^2 = \sum_{i=1}^{n} X_i^2 - n\,\overline{X}^2.$$

(a) Take the five observations $4,\ 7,\ 5,\ 8,\ 6$. Find $\overline{X}$. Then find S^2 in two ways: first from the definition, and then using the identity above.

(b) Let $X_1, \dots, X_n$ now be a random sample from the Bernoulli distribution, so that each observation is 0 or 1. Explain why $\sum_{i=1}^{n} X_i^2 = n\,\overline{X}$. Then use the identity above to show that

$$S^2 = \frac{n}{n-1}\,\overline{X}\,\big(1 - \overline{X}\big).$$

(c) Part (b) shows that for a sample of zeros and ones, the sample mean fixes the sample variance. Is this also true for a sample from some other population?

9. Let Z_1, Z_2, Z_3, Z_4, Z_5 be independent standard normal random variables. For each of the following, name the distribution and give the degrees of freedom. Say which definition you have used.

(a) $Z_1^2 + Z_2^2$

(b) $\dfrac{Z_1}{\sqrt{(Z_2^2 + Z_3^2)/2}}$

(c) $\dfrac{(Z_1^2 + Z_2^2)/2}{(Z_3^2 + Z_4^2 + Z_5^2)/3}$

(d) $\dfrac{Z_1^2}{(Z_2^2 + Z_3^2 + Z_4^2)/3}$

(e) Let T have the t distribution with k degrees of freedom. Show that T^2 has the F distribution with 1 and k degrees of freedom.

10. Let $X_1, \dots, X_{100}$ be a random sample from a population with mean 8 and variance 4. The shape of the population distribution is not known.

(a) Write down the mean and the variance of $\overline{X}$. Do the same for $\sum_{i=1}^{100} X_i$.

(b) Use the central limit theorem to find an approximate value of $P(\overline{X} > 8.3)$.

(c) Use the central limit theorem to find an approximate value of $P\big(\sum_{i=1}^{100} X_i \le 830\big)$.

(d) Parts (b) and (c) use the same entry of the table. Explain why. Write each of the two standardised quantities in terms of the other.

11. Let Y have the binomial distribution with $n = 400$ and $p = 1/2$.

(a) Write down $E[Y]$ and $\mathrm{Var}(Y)$, using Question 7.

(b) Find an approximate value of $P(Y \geq 215)$ by the central limit theorem. Do this first without the continuity correction, and then with it.

(c) The exact value of $P(Y \geq 215)$ is 0.0735. Which of your two answers is closer?

(d) Now let Y have the binomial distribution with $n = 30$ and $p = 0.01$. Find $E[Y]$ and the standard deviation of Y.

(e) Using the binomial formula, compute $P(Y = 0)$ and $P(Y = 1)$. Then find the exact value of $P(Y \geq 2)$.

(f) Find an approximate value of $P(Y \geq 2)$ by the central limit theorem, with the continuity correction. Compare it with your exact answer in part (e). By what factor do the two differ?

(g) The normal curve you used in part (f) has mean 0.3 and standard deviation 0.545. What probability does this curve give to values below -0.5? But Y is a count, so it is never negative. What does this say about the approximation?

(h) People often say that $n = 30$ is large enough for the central limit theorem. Here n is exactly 30. Looking at parts (d) to (g), which quantity seems to decide whether the approximation is good: n alone, or $np(1 - p)$?

How to present your work. Every answer on this sheet can be read from Tables A1 to A4 in the appendix of Ross. Some of them also need one of the two identities of Part A. You will not have to interpolate between two entries of a table anywhere. Each time you use a table, write down which table you used and which entry you read, and not only the final number. Give probabilities to four decimal places, as the tables do. A number written alone, with no sentence saying what it means, will not get full marks.